

CORTEX

Mobile Recon Framework

SECURITY ASSESSMENT REPORT

DVIA-v2

com.hightitudehacks.DVIAswiftv2

Platform	IOS
Scan Date	24 June 2026
Filename	DVIA-v2.ipa
Scan ID	476cbe33-5552-48...

0

CRITICAL

7

HIGH

2

MEDIUM

5

LOW

2

INFO

Prepared by

ferv

Static analysis operator

Executive Summary

Overall Risk Assessment: **HIGH RISK**

This report presents the findings from a static security analysis of the ios application **DVIA-v2** using Beetle Mobile Recon Framework. The analysis identified **16 total issues** across 16 findings, including **0 critical** and **7 high severity** items requiring immediate attention. Additionally, **0 potential hardcoded secrets** were detected in the app bundle.

Severity	Count	Description
HIGH	7	Significant security risk. Should be addressed promptly.
MEDIUM	2	Moderate risk. Address in next release cycle.
LOW	5	Minor issue. Address as part of normal maintenance.
INFO	2	Informational finding. No direct security impact.

CISO Summary

Overall Risk: **High** Security Grade: **C** MASVS Maturity: **weak** (35/100)

Security grade C (Fair). 0 critical and 7 high-severity findings. weak MASVS maturity (35/100 coverage).

Most Critical Issue: MD5 Hash Function Used

Business Risks

Risk	Why it matters
Weak data protection	Cryptography or local-storage weaknesses reduce confidentiality of data held on the device.

Prioritized Remediation

#	Item	Action
P1	MD5 Hash Function Used	Use SHA-256 via CryptoKit (iOS 13+) or CommonCrypto CC_SHA256.
P1	Keychain Item Without Access Control	Use kSecAttrAccessibleWhenUnlocked or kSecAttrAccessibleWhenPasscodeSetThisDeviceOnly for sensitive credentials.
P1	Provisioning Profile Expired	Re-sign the app with a valid provisioning profile before distribution.
P1	Mach-O Binary Not Position-Independent (PIE)	Rebuild with -Wl,-pie and ensure MACH_O_TYPE is MH_EXECUTE with MH_PIE set.

Application Information

Bundle ID	com.hightitudehacks.DVIAswiftv2
App Name	DVIA-v2
Version	2.0 (build 1)
Min iOS	12.0
File Size	9.2 MB
SHA-256	dabf92a5ca1cc00221fa3a12f1b58f10...
Framework	Native

Application Permissions (2 total)

Permission	Status	Description
NSCameraUsageDescription	NORMAL	Camera access
NSFaceIDUsageDescription	NORMAL	Face ID biometric

Developer Remediation Guide

Findings grouped by engineering area. Each group lists what was found, why it is dangerous, and how to fix it.

Crypto [P1] · 3 issue(s) · MASVS-CRYPTO-1

Found: MD5 Hash Function Used (DVIA-v2:18696); SHA-1 Hash Function Used (DVIA-v2:18695); Realm Database Without Encryption (DVIA-v2:100)

Why dangerous: A broken primitive or hardcoded key means the protected data can be recovered or forged by an attacker who obtains it.

Fix: Use AES-GCM with a Keystore-backed key, SHA-256+, and a CSPRNG; never hardcode keys/IVs.

```
KeyGenParameterSpec spec = new KeyGenParameterSpec.Builder(alias,
KeyProperties.PURPOSE_ENCRYPT|KeyProperties.PURPOSE_DECRYPT)
.setBlockModes(KeyProperties.BLOCK_MODE_GCM)
.setEncryptionPaddings(KeyProperties.ENCRYPTION_PADDING_NONE).build();
```

Storage [P1] · 1 issue(s) · MASVS-STORAGE-1

Found: Keychain Item Without Access Control (Parse:4403)

Why dangerous: An attacker who reaches this code path can leverage the weakness toward data access or control.

Fix: Review the evidence, confirm exploitability, and apply the platform's secure-coding guidance.

```
// See the remediation summary and references for the secure pattern.
```

MASVS Posture

Overall MASVS coverage: **35/100** (weak). Weakest area: **MASVS-CODE**. 2 positive control(s) detected.

Category	Coverage	Maturity	Missing Controls
MASVS-STORAGE	19/100	Weak	Encrypted Storage, No Sensitive Data in External Storage
MASVS-CRYPTO	39/100	Weak	Strong Cryptography, No Hardcoded Keys
MASVS-AUTH	40/100	Weak	Biometric Authentication, Secure Credential Handling
MASVS-NETWORK	40/100	Weak	Network Security Config, Certificate Pinning, No Cleartext Traffic
MASVS-PLATFORM	40/100	Weak	Safe WebView Configuration, Protected Components, Validated Deep Links
MASVS-CODE	0/100	Weak	Safe Input Handling, Up-to-date Dependencies
MASVS-RESILIENCE	60/100	Moderate	Integrity / Attestation, Strong App Signing
MASVS-PRIVACY	40/100	Weak	Minimal Permissions, Data Minimization

Strong controls: Keystore/Keychain-backed Keys, Root/Tamper Detection

Discovered Endpoints (51)

URL
http://%@.couchbase./
http://cacerts.digicert.com/DigiCertSHA2HighAssuranceServerCA.crt0
http://certs.apple.com/wwdr3.der01
http://crl.apple.com/aipca.crl0
http://crl.apple.com/root.crl0
http://crl3.digicert.com/sha2-ha-server-g4.crl04
http://crl4.digicert.com/sha2-ha-server-g4.crl0L
http://damnvulnerableiosapp.com
http://highaltitudehacks.com
http://highaltitudehacks.com/2013/06/16/ios-application-security-part-3-understanding-the-objective-c-runtime
http://highaltitudehacks.com/2013/07/02/ios-aio-application-security-part-4-runtime-analysis-using-cycript-yahoo-weather-app

http://highaltitudehacks.com/2013/07/02/ios-application-security-part-5-advanced-runtime-analysis-and-manipulation-using-cycript-yahoo-weather-app
http://highaltitudehacks.com/2013/07/25/ios-application-security-part-8-method-swizzling-using-cycript
http://highaltitudehacks.com/2013/08/20/ios-application-security-part-11-analyzing-network-traffic-over-http-slash-https
http://highaltitudehacks.com/2013/09/17/ios-application-security-part-16-runtime-analysis-of-ios-applications-using-inalyzer
http://highaltitudehacks.com/2013/10/26/ios-application-security-part-20-local-data-storage-NSUserDefaults
http://highaltitudehacks.com/2013/11/08/ios-application-security-part-21-arm-and-gdb-basics
http://highaltitudehacks.com/2013/12/17/ios-application-security-part-22-runtime-analysis-and-manipulation-using-gdb
http://highaltitudehacks.com/2013/12/17/ios-application-security-part-24-jailbreak-detection-and-evasion
http://highaltitudehacks.com/2013/12/17/ios-application-security-part-26-patching-ios-applications-using-ida-pro-and-hex-fiend
http://highaltitudehacks.com/2014/01/17/ios-application-security-part-28-patching-ios-application-with-hopper
http://highaltitudehacks.com/2014/03/07/ios-application-security-part-30-attacking-url-schemes
http://ocsp.apple.com/ocsp03-aipca070/
http://ocsp.apple.com/ocsp03-applerootca0.
http://ocsp.apple.com/ocsp03-wwdrg3040
http://ocsp.digicert.com0M
http://www.apple.com/DTDs/PropertyList-1.0.dtd
http://www.google-analytics.com/batch
http://www.google-analytics.com/collect
http://www.thejuniperfund.org/
https://api.login.yahoo.com/oauth2/device_session
https://api.mixpanel.com/track/?data=%&ip=1
https://api.parse.com/1
https://cfg.flurry.com/sdk/v1/config
https://data.flurry.com/v1/flr.do
https://github.com/CocoaLumberjack/CocoaLumberjack/wiki/CustomFormatters
https://github.com/parse-community/Parse-SDK-iOS-OSX/issues/new

https://github.com/yapstudios/YapDatabase/wiki/Views
https://realm.io/docs/objc/latest
https://realm.io/docs/objc/latest/#to-many
https://realm.io/docs/objc/latest/api/Classes/RLMObject.html
https://realm.io/docs/swift/latest/api/Classes/Object.html
https://ssl.google-analytics.com/batch
https://ssl.google-analytics.com/collect
https://twitter.com/prateekg147
https://www.apple.com/appleca/0
https://www.apple.com/certificateauthority/0
https://www.digicert.com/CPS0
https://www.google.com
https://www.youtube.com/watch?v=6dHXcoF590E
https://www.youtube.com/watch?v=HsV6jaA5J2I

Attack Surface Map

Custom URL Schemes

Scheme
dvia://
dviaswift://

Third-Party SDKs (4 detected)

SDK Name	Category	Package Prefix
Bolts	unknown	
Parse	unknown	
Realm	unknown	
RealmSwift	unknown	

Vulnerable Components (4 detected · 0 CVEs)

Component	Version	Ecosystem	CVEs	KEV	Worst
Bolts	1.9.1	CocoaPods	0	0	-
Parse	1.19.4	CocoaPods	0	0	-

Realm	4.1.1	CocoaPods	0	0	-
RealmSwift	4.1.1	CocoaPods	0	0	-

Source: OSV.dev (cached 24h) + CISA KEV feed.

Security Score

C

67/100

Fair

High Risk

Moderate security issues detected. Several items require attention.

Severity	Findings	Points/item	Total Deducted
HIGH	7	8	-24
MEDIUM	2	3	-6
LOW	5	1	-3

String Analysis (1 categories)

Category	Severity	Count	Sample Values
Hidden URL / API Endpoint	INFO	3	https://github.com/CocoaLumberjack/CocoaLumberjack/wiki/Cust...